TraceIQ Beginner Learning & Architecture Guide

Welcome! This documentation series is specifically created so you can **understand every concept, architecture decision, and code structure** behind TraceIQ from first principles.

Whether you are new to Retrieval-Augmented Generation (RAG), vector databases, or microservice architecture, this guide explains each component step-by-step.

Learning Roadmap

Guide	Description	What You Will Learn
01. What is RAG & TraceIQ?	The fundamentals of RAG and why we are building TraceIQ	LLM limitations, what RAG is, why simple RAG fails in engineering, and the 5 multi-source evidence types.
02. Architecture & Technical Deep Dive	Technical design & algorithms	MongoDB Vector Search, Full-Text Search, Hybrid RRF Search, Embeddings, and LangChain orchestration.
03. Codebase Tour & File Guide	Tour of this repository	What every folder and file in `backend/` and `frontend/` does and how they interact.
04. The 4 Judge Evaluation Scenarios	Hackathon benchmark scenarios	The 4 specific challenges judges will test: Cross-source synthesis, Exact keyword lookup, Contradictory evidence, and Abstention.

How to Use These Documents

1. **Read in order**: Start with Guide 01 and progress sequentially.
2. **Experiment alongside**: Refer back to the codebase files linked throughout the text.
3. **Ask questions**: Every time we complete a phase or feature, we will discuss the concepts behind it before moving to the next.